

Wuyang Chen

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Date of Birth: 2003.06.02 | Native Place: Chuzhou, Anhui

EDUCATION (EDUCATION)

- **City University of Hong Kong** M.Sc. in Artificial Intelligence GPA: 3.66/4.3 2025.09 - 2026.10
– **Core Courses:** Vision and Image (A), Intelligent Systems (A), Dynamic Programming and Reinforcement Learning (A-)
- **Beijing Jiaotong University** B.Eng. in Computer Science and Technology GPA: 86.6/100 2021.09 - 2025.06
– **Core Courses:** C++ (97), Mathematical Analysis (100), Higher Algebra (93), Deep Learning (90), Python (A), Data Structures (95)

PROFESSIONAL SKILLS (SKILLS)

- **Development & Simulation:** Python/C++, ROS (Noetic) Middleware & Distributed Node Development, MuJoCo, Unity, Sim2Real
- **Algorithms & Theory:** Modern Control Theory (NLESO, MDP, Value Iteration), PyTorch, High-dimensional Sequential Data (LSTM), Environment Perception (U-Net), Trajectory Prediction (Dynamic/Multi-object Prediction)

RESEARCH & PROJECTS (RESEARCH)

- **Path Planning and Motion Control of Autonomous Underwater Robots (AUR)** 2026.01 - Present
Tech: Receding Horizon Planning, Nonlinear ESO, Marine Dynamics, MuJoCo, ROS [ Github]
– **Project Background:** Designed and implemented a global-local cascaded "Planning-Guidance-Control" closed-loop system for complex maritime navigation. The project is nearing completion, with plans to formalize and publish a paper with the advisor.
– **Global Architecture:** The upper layer utilizes topological waypoints for collision-free global guidance to avoid local minima deadlocks; the lower layer implements the Fossen 3-DOF underactuated nonlinear dynamics model to resolve mismatches between geometric planning and physical execution constraints.
– **Trajectory Optimization:** Designed an RHP online planning module incorporating a piecewise polynomial flexible repulsive potential field, providing a smooth escape gradient while maintaining collision avoidance baselines. Combined tracking errors and heading constraints to achieve smooth switching between complex dynamic obstacle avoidance and high-precision tracking.
– **Control & Disturbance Rejection:** Applied NLESO in the control layer to observe and feedforward-compensate "lumped disturbances" (time-varying currents and unmodeled damping) in real-time; introduced a Reference Governor with Jerk saturation constraints to dynamically shape upper-level step commands using motor torque, eliminating high-frequency actuator oscillations.
– **Simulation Deployment:** Reconstructed the distributed system based on ROS Noetic and built an "Algorithm-ROS-MuJoCo" closed-loop co-simulation testing platform, achieving high-fidelity dynamics verification. Currently pushing for Sim2Real deployment in physical environments.
- **Multi-Object Hybrid Tracking and Future Dynamic Prediction System** 2026.02 - 2026.04
Tech: Sequential Deep Learning, Voxelization, LSTM, Deep Learning + Physics Model Fusion [ Github]
– **Perception & Tracking:** Processed real-world high-dimensional LiDAR point cloud data (~35k points/frame). Designed a 3D object detection pipeline based on 0.2m high-precision Voxelization, RANSAC, and DBSCAN clustering, reducing perception false positives from 10688 to 6800. Integrated a 6D Kalman Filter tracker to improve the MOTA metric from a baseline of -0.6479 to -0.3858.
– **Sequential Modeling & Prediction:** Addressed nonlinear turn drift for agents in complex environments by building a high-precision kinematic baseline based on constant turn rate and velocity. Utilized an LSTM architecture to encode high-dimensional history trajectories to capture nonlinear evolution patterns.
– **Performance Breakthrough:** Overcame the tangential divergence effect of traditional models in long-term prediction. In systematic evaluations, the overall average L2 prediction error dropped from 3.74m to 2.64m; under a challenging 5.5s prediction window, the L2 error was reduced from 4.43m to 3.25m.

- **Hypergraph Learning Framework based on Multi-Omics**
Tech: Multi-Omics, Hypergraph Modeling, Graph Neural Networks, Dimensionality Reduction
– Designed a hypergraph learning framework based on biological multi-omics data, breaking through the limitations of traditional linear methods in processing high-order data.

2025.09 - 2025.12

INTERNSHIPS (INTERNSHIP)

- **Institute of Software, Chinese Academy of Sciences**
– Participated in the ALS machine recognition challenge, specifically trained YOLOv5 networks, and secured 3rd place in the group.
– Developed dataset annotation tools and constructed data processing workflows.
– Wrote handwriting recognition algorithms involving pixel aggregation, linkage models, and branch-and-bound strategies.
 - **Tencent Technology (Shenzhen) Co., Ltd.**
– **Tech Stack:** Flutter Architecture, Routing Navigation, Protocol Communication, Android
 - **4399 Network Co., Ltd.**
– **Tech Stack:** Unity Engine, MVC Architecture, Front-end and TypeScript

AI & Data Science Intern

2024.04 - 2024.11

Front-end Development Intern

2025.08 - 2025.09

Client Development Intern

2025.04 - 2025.05

HONORS & ENGLISH (HONORS & ENGLISH)

- **National Scholarship, First-Class Academic Scholarship, "Merit Student" Honor**
 - **IELTS 6.5, GRE 319+3.5 (Writing), Passed CET-4/6**

2022.12, etc.

2024.12, etc.